

Forecasting Norwegian Salmon Export Prices: A Time-Series Analysis Using VAR, ARIMA and ETS

Torjus Arne Ljungqvist (S161489)
Jan Henrik Haudemann-Andersen (S162647)
Sebastian Nes Romero (S185875)

Predictive Analytics (CDSCO1005E)
MSc. Business Administration and Data Science
Copenhagen Business School
Spring 2026
Pages: 12
Characters: 33,690

Contents

1	Introduction	2
2	Related Work	2
3	Data Description	3
4	Data Exploration and Preparation	3
4.1	Preparation	3
4.2	Temporal Patterns and Mathematical Transformation	3
4.3	Stationarity and Differencing	5
4.4	Structural Breaks	6
5	Forecasting Models	6
5.1	Model Choice	6
5.2	ARIMA	7
5.3	ETS	8
5.4	VAR	9
6	Forecast Accuracy and Final Forecast	10
6.1	Accuracy Comparison	10
6.2	Final 12-Month Forecast	11
7	Conclusion and Recommendations	12
	References	13
	Appendices	14
A	Seasonal and Subseries Plots	14
B	Stationarity Tests	15
C	Additional Diagnostic Figures	16
D	Model Selection	17
E	Residual Diagnostics	18
F	Forecast Figures	20

1. Introduction

Norwegian salmon is one of the most traded food commodities in the world, yet its price is notoriously difficult to forecast. Driven by biological production cycles that span two to three years, a spot market that clears weekly, and demand shocks capable of moving prices by 30% or more within months, salmon export prices combine the volatility of financial assets with the supply rigidity of agricultural commodities (Asche et al., 2016). This makes accurate price forecasting both commercially valuable and genuinely challenging. In 2024, Norway exported over 1.25 million tonnes of salmon worth NOK 122.9 billion, accounting for 70 percent of total Norwegian seafood export value (Norwegian Seafood Council, 2025).

The commercial stakes of getting the forecast right are considerable. Producers must allocate harvest volumes months in advance, exporters price forward contracts against an uncertain future spot price, and buyers plan procurement budgets for annual cycles. A 12-month forecast horizon is particularly relevant as it aligns with typical planning and contracting cycles in the industry. Yet as this paper demonstrates, even the best-specified statistical models leave a large share of price variation unexplained, and prediction intervals widen rapidly beyond the first few months, an honest reflection of how much genuine uncertainty surrounds salmon prices at medium-term horizons (Bloznelis, 2018).

This paper forecasts the monthly export price of Norwegian fresh and chilled salmon one year ahead using three time series models: Exponential Smoothing (ETS), AutoRegressive Integrated Moving Average (ARIMA), and Vector Autoregression (VAR). The dataset consists of weekly observations from Statistics Norway (SSB) covering January 2000 to April 2026, aggregated to monthly frequency for modeling purposes, and includes both export price (NOK/kg) and export volume (tonnes). The VAR model treats price and volume as jointly endogenous, exploiting the dynamic relationship between the two series documented in Section 3. The primary research question is addressed as:

“Which time series model best predicts the monthly export price of Norwegian salmon over a 12-month horizon?”

Generative AI (Claude) was used during the writing process to review the R code, suggest structural and stylistic improvements to the manuscript, and check internal consistency. All analytical choices, model specifications, results, and final text are the authors’ own. No GenAI-generated text or figures have been inserted into the paper.

2. Related Work

The question of whether statistical models can usefully forecast salmon prices has been investigated since at least the late 1990s, with the literature converging on a consistent finding: structured time series models do capture meaningful patterns in salmon price dynamics, but gains over simple benchmarks are modest. Guttormsen (1999) was among the first to demonstrate this, applying ARIMA, VAR, and Holt-Winters exponential smoothing to weekly farmed Atlantic salmon prices and showing that all three approaches could extract predictable structure from the series. The methodological foundation established by that study, univariate and multivariate time series models estimated on Norwegian export data, directly informs the present paper.

Nearly two decades later, Bloznelis (2018) benchmarked sixteen forecasting methods against weekly Norwegian salmon spot prices, including classical time series models, machine learning techniques, and futures prices. The headline result was paradoxical: nominal accuracy gains over a naïve benchmark were small, yet the economic value of the forecasts was considerable. This finding reframes what counts as a useful salmon price forecast, marginal improvements in point accuracy translate into meaningful commercial value, which is precisely the motivation for the 12-month forecasting exercise undertaken here.

The most directly relevant precedent is Tharmarajah and Gjesdal (2020), who compared univariate and multivariate methods for Atlantic salmon price over a 12-month horizon and found that simpler univariate models outperformed more complex multivariate approaches. Rather than

settling the question, this finding motivates the present study: by applying the same model classes to a longer and more recent dataset spanning 2000 to 2026, we test whether the univariate advantage holds across a sample that includes the 2015 regime shift, the COVID-19 disruption, and the subsequent price peak of 2023.

3. Data Description

The data are obtained directly from Statistics Norway (SSB) via their public API, table 03024, which reports weekly salmon export statistics (Statistics Norway, 2026). The dataset covers January 2000 to April 2026 and contains two variables: export price (NOK/kg) and export volume (tonnes) for fresh and chilled farmed Atlantic salmon. Fresh and chilled salmon is selected over frozen, as it accounts for approximately 95% of total salmon export value across the sample period, making it the dominant and most policy-relevant category for price forecasting (Norwegian Seafood Council, 2025). **All prices are in nominal NOK, meaning they are not adjusted for inflation. This is appropriate because the objective is to forecast the actual export price faced by market participants, rather than the real purchasing-power-adjusted price.** The dataset consists of 1,375 weekly observations, which are aggregated to monthly frequency for modeling purposes. Volume is summed and price is averaged within each month, yielding 316 monthly observations. The data are not seasonally adjusted and are reported as FOB values, meaning transport costs and insurance are included but duties and VAT are excluded (Statistics Norway, 2026).

The export price variable is the primary forecasting target, measured as the average price per kilogram in Norwegian kroner for the given month. Export volume in tonnes is retained as a secondary variable and used jointly with price in the VAR model, where both series are treated as endogenous.

Both variables are included together because of how salmon prices are formed. Norwegian salmon trades on a weekly spot market where supply is largely predetermined by biological constraints: fish require two to three years from smolt to harvest, so producers cannot adjust production in response to short-term price signals. Individual

farms have limited harvest-timing flexibility but face continued feed expenses and constant slaughterhouse capacity (Asheim et al., 2011). Short-run salmon supply is therefore highly inelastic with respect to price.

This inelasticity implies a negative short-run relationship between price and volume: high supply pushes prices down, low supply raises them. The relationship reverses only in the long run as global demand has grown steadily over the sample (Norwegian Seafood Council, 2025). Because price and volume are jointly determined rather than one causing the other, a VAR treating both as endogenous is the natural framework (Sims, 1980), as discussed in Section 5.

4. Data Exploration and Preparation

4.1 Preparation

The raw SSB API output is reshaped from long to wide format with price (NOK/kg) and volume (tonnes) as separate columns. Weekly observations are aggregated to monthly frequency. Volume is summed within each calendar month and price is averaged. Aggregation reduces high-frequency noise inherent in weekly export statistics while preserving the trend and seasonal structure relevant for a 12-month forecast horizon. The Norwegian “U”-format week identifier provided by SSB is converted to ISO yearweek and then to yearmonth for compatibility with the **tsibble** framework. Observations with missing values in either variable are removed. The sample is truncated at April 2026. Since the SSB API updates weekly, the most recent calendar month in any given data pull is typically incomplete, meaning price averages and volume totals for that month are based on only a subset of weeks. Fixing the end date at April 2026 avoids the inclusion of such partial-month observations and ensures that the analysis is reproducible regardless of when the API is queried. The resulting **tsibble** contains 316 monthly observations spanning January 2000 to April 2026.

4.2 Temporal Patterns and Mathematical Transformation

Figure 1 shows the time plot of the raw and log-transformed price series. The raw series reveals a clear stochastic upward trend, rising from approximately NOK 33/kg in January 2000 to peaks

above NOK 120/kg in 2023 before a partial decline. Two distinct regime shifts are visible. The first occurs around 2014–2015, when Russia’s import ban on Norwegian seafood simultaneously reduced export volume and redirected supply toward European and Asian markets, driving a sustained upward shift in the price level (Asche et al., 2016; Norwegian Seafood Council, 2025). The second begins around 2020, where pandemic-related demand disruptions and subsequent supply chain pressure generate a sharp increase in both the level and variance of prices. The pronounced fanning of the series from 2020 onwards indicates heteroskedasticity. Price fluctuations grow larger as the level rises, violating the constant-variance assumption required for modeling on the raw scale.

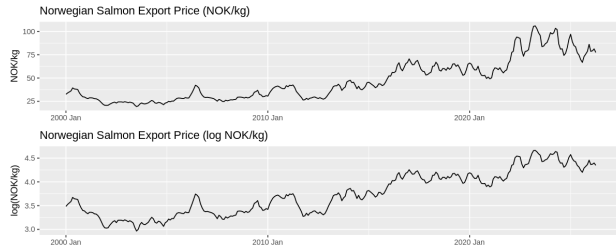


Figure 1: Norwegian salmon export price in NOK/kg (top) and log NOK/kg (bottom), January 2000 – April 2026.

To stabilize the variance, a log transformation is applied. The Guerrero method returns an optimal Box-Cox lambda of **0.049**, which is indistinguishable from zero and confirms that the log transformation is the statistically appropriate choice (Hyndman & Athanasopoulos, 2021, ch. 3). We prefer the log over the mechanical Box-Cox optimum for a second reason: changes in the log-transformed series correspond directly to monthly percentage changes in price, which is the natural and interpretable unit for commodity price analysis. As Figure 1 illustrates, the log transformation effectively removes the fanning pattern visible in the raw series, producing a more homoskedastic series suitable for time series modeling.

The seasonal and subseries plots (Figure 2) reveal a mild but consistent annual pattern across the sample. Prices tend to be above their annual average in Q1, dip during the summer months of June through August, and partially recover towards year-end. This pattern is consistent with

well-documented seasonal dynamics in Atlantic salmon markets, where summer supply increases from faster-growing fish in warmer waters coincide with subdued retail demand during holiday periods, depressing prices (Asche et al., 2016). The subseries plot confirms that the seasonal averages are stable across years, with no evidence of an evolving seasonal structure that would favour a time-varying seasonal specification.

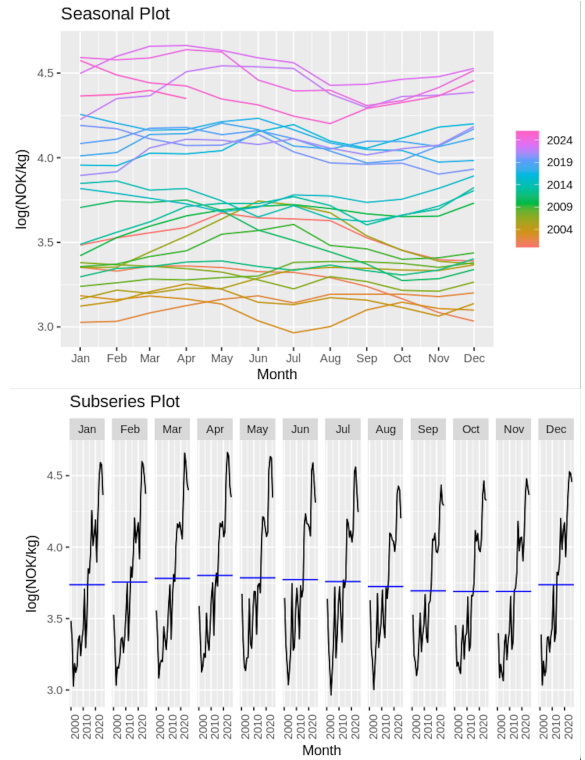


Figure 2: Seasonal plot (top) and subseries plot (bottom) of log salmon export price. The within-year dip in summer months is consistent across the full sample.

STL decomposition with a 13-month trend window and a fixed seasonal window (Figure 3) decomposes the log price series into its constituent components (Cleveland et al., 1990). The trend component is clearly dominant, accounting for the large majority of total variation in the series and capturing both the pre-2015 moderate growth phase and the steep post-2015 appreciation. The seasonal component is stable and relatively small in magnitude, approximately $\pm 3\%$ on the log scale, consistent with the visual impression from the seasonal plots. The remainder component is generally well-behaved but shows noticeably larger irregular fluctuations from 2020 onwards, reflecting the idiosyncratic volatility of the pandemic period that neither the trend nor the seasonal component can absorb. Together, the decomposition confirms that the series is dominated by a stochastic trend and that any modeling approach must account explicitly for non-stationarity.

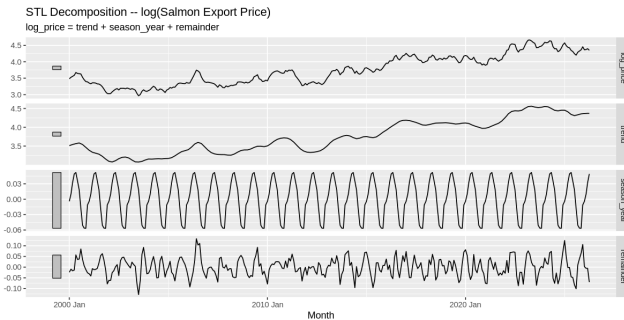


Figure 3: STL decomposition of log salmon export price. Trend window = 13 months, periodic seasonal window.

4.3 Stationarity and Differencing

Stationarity is assessed using the ADF and KPSS tests jointly, following Kwiatkowski et al. (1992). Using both tests reduces the risk of drawing incorrect conclusions from either test alone. The ADF tests the null of a unit root against a stationary alternative (Dickey & Fuller, 1979), while the KPSS tests the null of stationarity, making their combination more diagnostic than either test in isolation. Given the visible trend in Figure 1, both tests are initially specified to allow for a deterministic trend (ADF with type = “trend”, KPSS with type = “tau”).

Applied to log price in levels, the two tests produce conflicting results. The ADF tau3 statistic

of -4.066 exceeds the 1% critical value of -3.98 , formally rejecting the unit root null. However, this rejection should be treated with caution. The ADF test regression includes a highly significant time trend term ($t = 3.86$, $p < 0.001$), and in the presence of a strong deterministic trend the ADF can reject spuriously even when the series is not genuinely mean-reverting (Kwiatkowski et al., 1992). The joint F-statistics $\phi_2 = 6.105$ and $\phi_3 = 8.299$ narrow the confidence further, both rejecting at the 5% level but only marginally. The KPSS tau statistic of 0.345 tells a clearer story: it exceeds even the 1% critical value of 0.216 , strongly rejecting trend-stationarity. When ADF and KPSS both reject simultaneously, the standard interpretation is near-unit-root behaviour, where the series is close to but not cleanly inside either the stationary or unit-root region (Hyndman & Athanasopoulos, 2021). Given the unambiguous visual evidence of a trending, non-mean-reverting series in Figure 1, first differencing is applied as the conservative and standard remedy. In economic terms, this transforms the series from price levels, which drift unpredictably over time, to monthly percentage changes in price, which are stationary and form a stable basis for forecasting.

After applying one regular difference to log price, both tests agree. The KPSS statistic drops to 0.063 , well below the 10% critical value of 0.347 , failing to reject stationarity. The ADF statistic of -4.493 strongly rejects the unit root at the 1% level (critical value -2.58). Automated selection confirms that one regular difference is sufficient and that no seasonal differencing is required (Hyndman & Athanasopoulos, 2021), consistent with the small and stable seasonal component observed in the STL decomposition. The first-differenced log price series, corresponding to monthly percentage changes in price, is therefore stationary and forms the basis for all subsequent modeling.

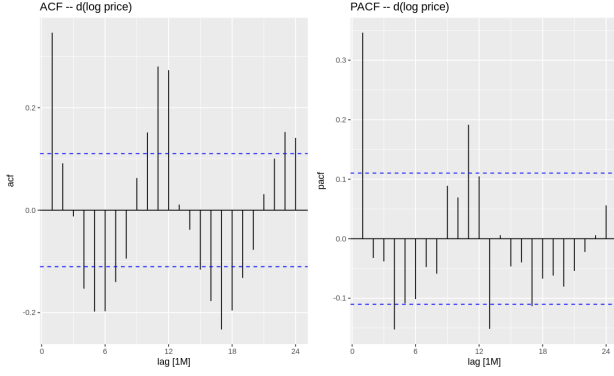


Figure 4: ACF and PACF of first-differenced log salmon export price.

4.4 Structural Breaks

The time plot in Figure 1 shows a pronounced level shift in salmon prices around 2014–2015, raising the question of whether the data-generating process underwent a structural change that would invalidate parameter estimates pooled across the full sample. The QLR (sup-F) test is applied to the first-differenced log price series with 15% trimming at each end (Andrews, 1993; Zeileis et al., 2002). The test is deliberately conducted on the differenced series, not the levels, because the relevant question for forecasting model stability concerns the dynamics, that is, whether the relationship between consecutive monthly changes is stable over time, rather than whether the price level itself shifted. A one-time level shift of the kind induced by the 2015 market shock is fully absorbed by first differencing and does not constitute a structural break in the dynamics.

Figure 5 shows the F-statistics across all candidate break dates. The test statistic never approaches the upper boundary corresponding to the 5% critical value, and the formal sup-F p-value is 0.623. We therefore fail to reject the null of parameter stability across the full sample. The well-documented 2014–2015 level shift is real and economically significant, but it manifests as a permanent shift in the price level rather than a change in how price evolves month-to-month. Retaining the full 316-month sample preserves substantially more observations for model estimation and is statistically justified by this result.

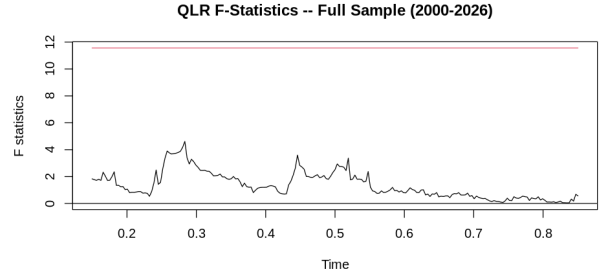


Figure 5: QLR F-statistics for the first-differenced log price series, January 2000 – April 2026. The horizontal line indicates the 5% critical value. The test statistic never crosses the threshold (sup-F $p = 0.623$).

5. Forecasting Models

5.1 Model Choice

Three forecasting models are estimated on the training sample (January 2000 to December 2022) and evaluated on a hold-out test set (January 2023 to April 2026). ARIMA is the natural choice for this series given that first differencing was required to achieve stationarity, and the ACF and PACF of the differenced log price series reveal clear autocorrelation structure at both short and seasonal lags, indicating that past price observations contain predictive information about future prices (Box & Jenkins, 1970). ETS is included as an alternative that handles trend and seasonality through exponential smoothing rather than differencing, making no assumption about the order of integration (Hyndman et al., 2008). Where ARIMA explicitly models autocorrelation structure, ETS captures the trend and seasonal patterns identified in the STL decomposition through adaptive weighting of past observations, making the two approaches methodologically complementary (Hyndman & Athanasopoulos, 2021). VAR is included to test whether jointly modeling price and export volume improves forecast accuracy over the univariate models (Sims, 1980). As established in Section 3, the joint determination of price and volume in the salmon spot market makes a VAR treating both variables as endogenous the natural statistical framework, a choice further supported by the Granger causality test (Granger, 1969), which confirms that past export volume significantly predicts price beyond price history alone ($F = 2.94$, $p = 0.001$ at lag order 11), and that price also Granger-causes volume

($F = 2.39$, $p = 0.008$). The bidirectional causality confirms that price and volume are genuinely interdependent, as past volume helps predict future prices, and past prices feed back into future volumes through harvest timing decisions, consistent with the market dynamics described in Section 3. However, it should be noted that the related literature reviewed in Section 2 has generally found that simpler univariate models outperform multivariate alternatives for salmon price forecasting (Tharmarajah & Gjesdal, 2020), which is why all three model classes are evaluated empirically.

5.2 ARIMA

Supply shocks in the salmon spot market (e.g. disease outbreaks reducing biomass) propagate through prices over several months until new cohorts reach harvest size, generating exactly the kind of autocorrelation structure ARIMA captures (Box & Jenkins, 1970). The ACF and PACF (Figure 4) of the first-differenced log price series confirm this empirically, with significant short-lag autocorrelation and a spike at lag 12 matching the annual seasonal pattern from the STL decomposition.

Based on the ACF and PACF, a manual candidate $\text{ARIMA}(0,1,1)(1,0,0)[12]$ is specified. **The manual specification includes only a single moving average term to capture short-run price momentum, but does not account for the additional autoregressive structure visible in the PACF at lags 1 and 2, meaning it relies solely on recent forecast errors rather than recent price levels to predict the next observation.** An automatic search using the Hyndman-Khandakar algorithm (Hyndman & Khandakar, 2008) selects $\text{ARIMA}(2,1,1)(1,0,0)[12]$ ($\text{AICc} = -883.7$), which adds two autoregressive terms that directly incorporate the last two months of price changes as predictors, giving the model a richer description of short-run price dynamics. This specification outperforms the manual candidate ($\text{AICc} = -879.0$) while retaining the same seasonal structure and is therefore preferred. Model selection is based on AICc rather than AIC to account for the finite-sample correction appropriate for this sample size (Hurvich & Tsai, 1989). The selected model uses two AR lags on monthly price changes plus an $\text{MA}(1)$ correction for recent forecast errors, with

a seasonal $\text{AR}(1)$ at lag 12 capturing the recurring annual price cycle driven by the within-year harvest and demand patterns documented in Section 3.

The Ljung-Box test at lag 24, adjusted for 4 estimated parameters, yields $p = 0.110$, failing to reject the null of white noise residuals (Ljung & Box, 1978). ARIMA is the only model among the three to pass residual diagnostics cleanly, suggesting it captures the autocorrelation structure of the salmon price series more completely than both ETS and VAR. Unlike ETS, which smooths over autocorrelation patterns rather than modeling them directly, ARIMA explicitly accounts for the short-term price dependencies visible in the ACF and PACF, making it better suited to a series where recent price momentum matters. The residual plots are shown in Figure 6.

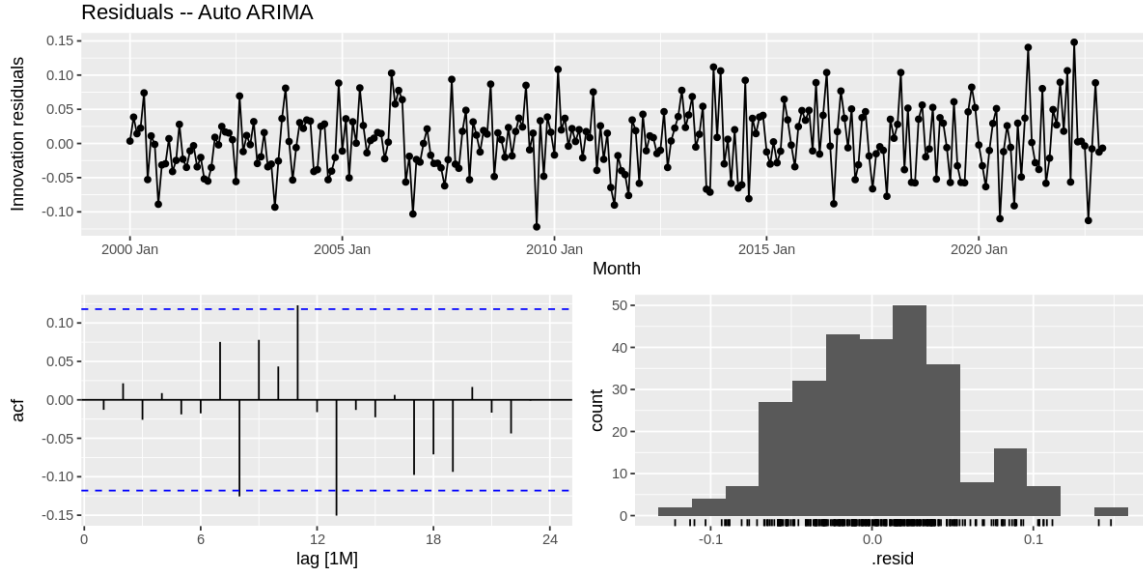


Figure 6: Residual diagnostics for ARIMA(2,1,1)(1,0,0)[12]. Residuals approximate white noise (Ljung-Box $p = 0.110$).

5.3 ETS

The STL decomposition revealed two features of the salmon price series that directly motivate the use of ETS: a dominant stochastic trend reflecting the long-run growth in global salmon demand, and a stable seasonal pattern with amplitude of approximately $\pm 3\%$ on the log scale driven by the within-year harvest cycle. ETS is designed precisely to track a slowly evolving level, trend, and seasonal component through adaptive smoothing, making it well suited to a series with these characteristics (Hyndman et al., 2008). Where ARIMA models the autocorrelation structure of the differenced series, ETS instead focuses on decomposing the level series into its trend and seasonal components and extrapolating them forward, offering a complementary perspective on how salmon prices evolve over time (Hyndman & Athanasopoulos, 2021).

ETS(A,A,A) with additive error, additive trend, and additive seasonality is appropriate here because the log transformation has stabilised variance, making additive rather than multiplicative errors reasonable, and the seasonal pattern identified in the STL decomposition is stable in amplitude over the full sample, supporting an additive seasonal component (Hyndman et al., 2008). The smoothing parameter $\alpha = 0.937$ updates the level almost entirely from the most recent observation,

consistent with a spot market where recent prices are far more informative than older ones. The near-zero β indicates an essentially stable trend, while $\gamma = 0.063$ allows the seasonal pattern to update slowly. The residual diagnostics are shown in Figure 7.

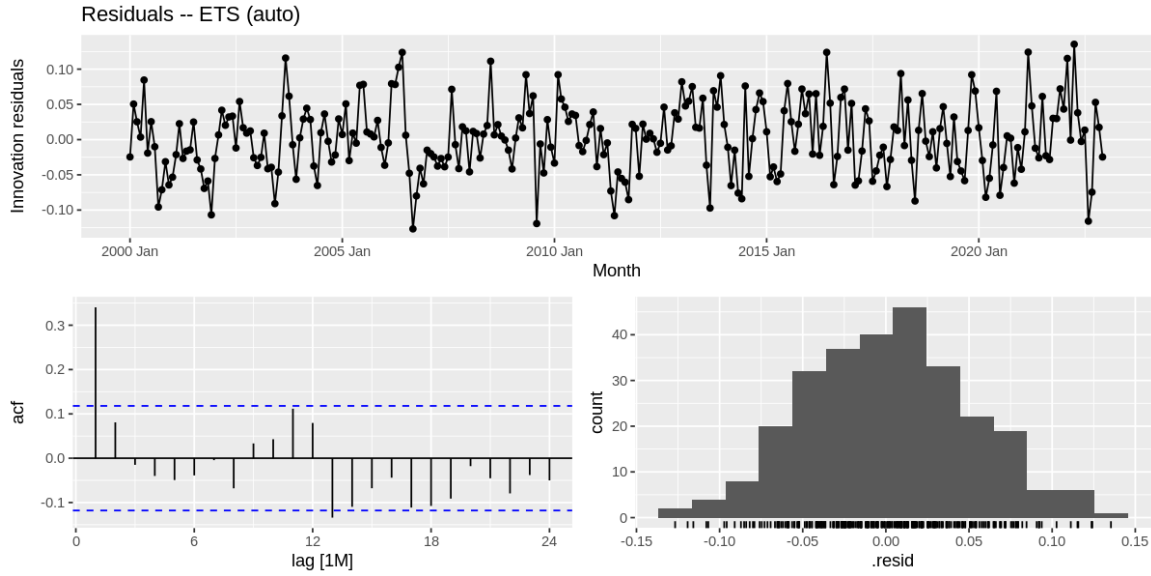


Figure 7: Residual diagnostics for ETS(A,A,A). Several significant ACF spikes indicate remaining autocorrelation (Ljung-Box $p = 4.7 \times 10^{-12}$).

The Ljung-Box test is evaluated across 24 lags to cover two full seasonal cycles, adjusted for the 17 estimated parameters of the ETS(A,A,A) model, reflecting the level, trend, seasonal smoothing weights and initial states. The test yields $p = 4.7 \times 10^{-12}$, strongly rejecting the null of white noise residuals (Ljung & Box, 1978). Despite capturing trend and seasonality, ETS does not fully account for the short-term autocorrelation structure in salmon prices, which is the same structure that the ACF and PACF revealed and that ARIMA explicitly models. This is a known limitation of exponential smoothing methods when applied to series with strong autocorrelation (Hyndman & Athanasopoulos, 2021), and is considered when interpreting forecast accuracy in Section 6, as the prediction intervals produced by ETS are unreliable despite the model achieving a competitive point forecast RMSE on the test set.

5.4 VAR

Having two jointly determined series, price and export volume, naturally raises the question of how to incorporate volume as a predictor. Dynamic regression would be one option, treating volume as an exogenous predictor of price. However, as established in Section 3, price and volume in the salmon market are not independent of each other. Volume affects price through supply

availability, but price can also influence harvest timing decisions at the farm level, meaning neither variable is cleanly exogenous to the other. This rules out dynamic regression, which imposes a one-way relationship, and makes VAR the appropriate framework (Sims, 1980). VAR treats both variables symmetrically as jointly endogenous, allowing each to be predicted by the history of both series simultaneously.

The VAR is estimated on the first differences of log price and log export volume, both differenced to remove the stochastic trends confirmed in Section 4. Lag order selection across a maximum of 24 lags selects $p = 11$ under the BIC criterion, preferred for parsimony (Schwarz, 1978). The high lag order is consistent with the biological production cycle of salmon documented in Section 3, where shocks to available supply propagate slowly through the market over many months before fully dissipating, requiring a longer memory structure than the univariate models. The price equation shows a significant own-lag at lag 1 (coefficient = 0.272, $p < 0.001$), while several volume lags contribute to the equation, consistent with the Granger causality result (Granger, 1969) showing that past volume predicts future price. The residual diagnostics are shown in Figure 8.

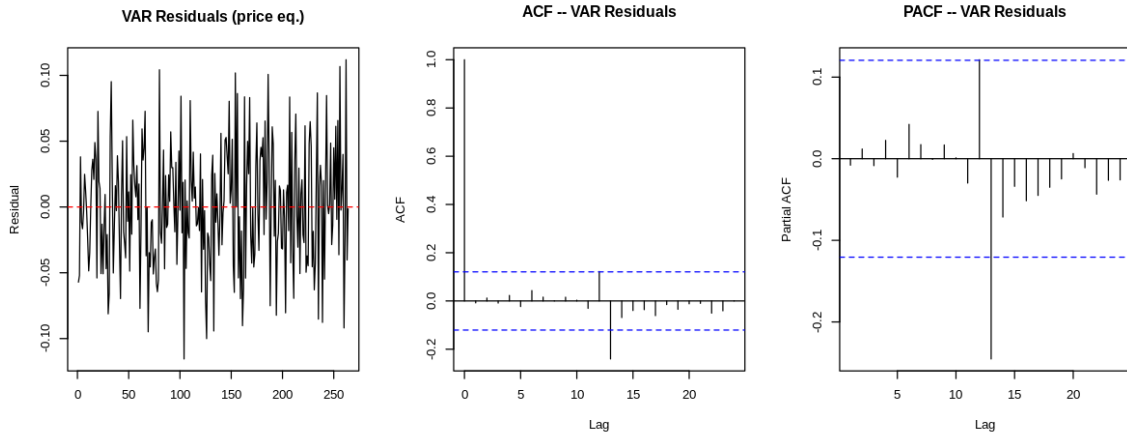


Figure 8: Residual diagnostics for VAR(11), price equation. Residuals show a significant spike at lag 12 and clustered post-2020 volatility (Portmanteau $\chi^2 = 85.8$, $df = 52$, $p = 0.002$).

The multivariate Portmanteau test at lag 24 rejects the null of no serial correlation ($\chi^2 = 85.8$, $df = 52$, $p = 0.002$), indicating the model does not fully capture the dynamics of the system. Despite the strong economic rationale and statistically significant Granger causality (Granger, 1969), the VAR struggles to produce a well-specified fit at monthly frequency, likely because the 11-lag structure consumes a substantial share of the 276 training observations, leaving limited degrees of freedom for precise parameter estimation (Lütkepohl, 2005). While the VAR incorporates information about export volume that the univariate models cannot use, this structural advantage does not translate into better residual diagnostics or forecast accuracy at the monthly horizon evaluated here, as shown in Section 6.

6. Forecast Accuracy and Final Forecast

6.1 Accuracy Comparison

Table 1 reports out-of-sample forecast accuracy for all three models on the 40-month hold-out test set (January 2023 to April 2026), evaluated on the log scale. All measures are computed on log-price to ensure comparability across models, as the VAR is estimated on the differenced log series and its forecasts are reconstructed by cumulating the predicted changes. RMSE and MAE are scale-dependent errors, while MASE is a scale-free measure defined relative to a seasonal naïve benchmark on the training set, with values below one indicating the model outperforms the bench-

mark (Hyndman & Koehler, 2006). The test-set forecasts shown in Figures 9–11.

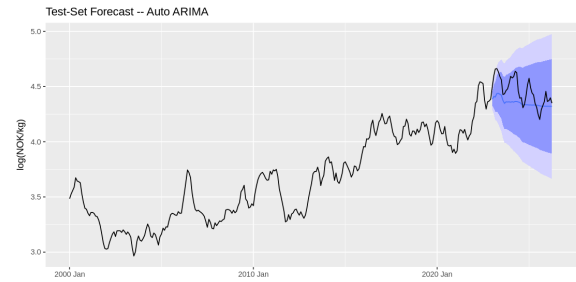


Figure 9: Test-set forecast, ARIMA(2,1,1)(1,0,0)[12], January 2023 – April 2026.

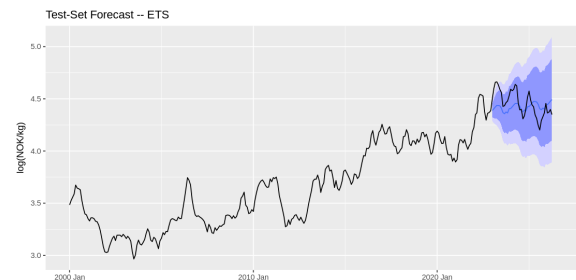


Figure 10: Test-set forecast, ETS(A,A,A), January 2023 – April 2026.

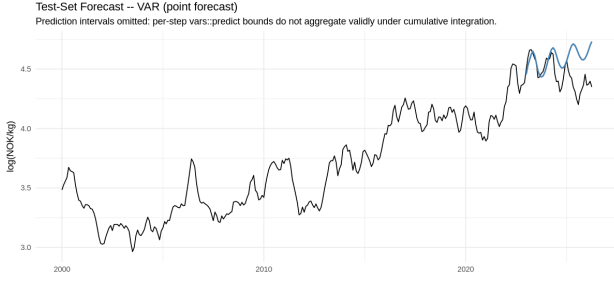


Figure 11: Test-set forecast, VAR(11) point forecast, January 2023 – April 2026.

Table 1: Out-of-sample forecast accuracy on log scale. MASE relative to seasonal naïve benchmark on training set (Hyndman & Koehler, 2006).

Model	RMSE	MAE	MASE
ETS(A,A,A)	0.129	0.112	0.657
ARIMA(2,1,1)(1,0,0)[12]	0.143	0.119	0.700
VAR(11)	0.202	0.153	0.902

ETS achieves the lowest RMSE and MAE on the test set, with ARIMA a close second. The RMSE gap of 0.014 on the log scale is roughly 1.4 percentage points in price terms — a narrow margin that may not be robust across alternative test periods. Both univariate models record MASE below one, meaning they outperform the seasonal naïve benchmark on out-of-sample prediction (Hyndman & Koehler, 2006). VAR performs substantially worse across all measures, with RMSE approximately 57 percent higher than ETS. This indicates that the additional complexity of the bivariate system does not translate into forecast accuracy gains at this horizon, a finding consistent with the related literature reviewed in Section 2, which found that simpler univariate models tend to outperform multivariate alternatives for salmon price forecasting (Tharmarajah & Gjesdal, 2020). Despite ETS’s lower point-forecast RMSE, ARIMA is selected for the final forecast. The decisive factor is residual diagnostics: ARIMA produces white noise residuals (Ljung-Box $p = 0.110$), while ETS fails this test severely ($p = 4.7 \times 10^{-12}$) (Ljung & Box, 1978). A model with autocorrelated residuals has not captured all predictable structure, rendering its prediction intervals unreliable even when point forecasts happen to be accurate (Hyndman & Athanasopoulos, 2021). For decisions like forward-contract pricing and procure-

ment planning, where interval reliability matters as much as the central estimate, ARIMA is the only model that delivers both.

6.2 Final 12-Month Forecast

The ARIMA(2,1,1)(1,0,0)[12] model is refitted on the full 316-month sample from January 2000 to April 2026 and used to generate a 12-month-ahead forecast from May 2026 to April 2027 (Figure 12). Refitting on the full sample makes use of all available information, including the most recent observations, and is the standard practice for producing a final operational forecast after model selection has been completed on the training set (Hyndman & Athanasopoulos, 2021).

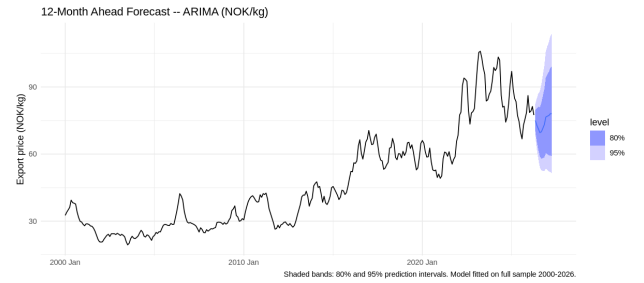


Figure 12: 12-month ARIMA forecast of Norwegian salmon export price (NOK/kg), May 2026 – April 2027, with 80% and 95% prediction intervals. Model fitted on full 316-month sample.

Table 2: 12-month salmon export price forecast (NOK/kg), May 2026 – April 2027.

Month	Forecast	L80	U80	L95	U95
2026 May	75.0	70.3	79.8	68.0	82.5
2026 Jun	72.8	65.4	80.5	61.9	85.0
2026 Jul	71.3	62.0	81.1	57.7	87.1
2026 Aug	69.6	58.9	80.9	54.1	88.0
2026 Sep	70.0	57.9	82.9	52.7	91.1
2026 Oct	71.5	58.1	86.0	52.4	95.4
2026 Nov	73.2	58.6	89.1	52.4	99.6
2026 Dec	76.5	60.3	94.2	53.6	106.0
2027 Jan	76.9	59.9	95.6	52.9	108.2
2027 Feb	77.2	59.4	96.8	52.3	110.1
2027 Mar	78.0	59.4	98.5	52.0	112.6
2027 Apr	78.1	59.0	99.3	51.4	113.9

Table 2 presents the monthly point forecasts and prediction intervals in NOK/kg, back-transformed from the log scale. The model forecasts salmon export prices in the range of NOK 70–78 per kilogram over the forecast horizon, implying a modest

continuation of the partial price recovery observed in late 2025 and early 2026 following the decline from the 2023 peak. The forecast path reflects the seasonal structure captured by the SAR(1)[12] component: prices are expected to dip slightly in the summer months before partially recovering towards year-end, consistent with the recurring within-year pattern documented in Section 4 (Asche et al., 2016).

The prediction intervals widen substantially over the forecast horizon. At one month ahead the 95% interval spans approximately ± 7 NOK/kg around the point forecast, but by month 12 the interval has widened to roughly ± 28 NOK/kg. This rapid widening reflects the genuine difficulty of medium-term salmon price forecasting. The model captures trend, seasonality, and short-run autocorrelation, but a large share of monthly variation is driven by factors outside its information set — disease outbreaks, exchange rate movements, and unexpected demand shifts in key markets. These compound over the forecast horizon, and the widening intervals reflect what time series models alone can deliver rather than a limitation of the specific model chosen (Hyndman & Athanassopoulos, 2021).

7. Conclusion and Recommendations

Of the three models evaluated, ARIMA is the recommended choice. ETS produces marginally lower test-set RMSE but leaves substantial residual autocorrelation, rendering its prediction intervals unreliable. VAR performs worst on all accuracy measures (RMSE $\sim 57\%$ higher than ETS), likely because its 11-lag structure consumes too many of the 276 training observations for precise parameter estimation. ARIMA is the only model whose intervals can be trusted, widening from approximately NOK 7/kg at one month ahead to roughly NOK 31/kg at twelve months. For decisions that depend on quantifying price risk, interval reliability matters as much as the point forecast, and ARIMA is the only model that delivers both.

Based on this model, Norwegian salmon export prices are forecast in the range of NOK 70–78 per kilogram from May 2026 through April 2027, with a modest dip over the summer months before a partial recovery towards year-end. This seasonal pattern is consistent across all 26 years of data and

is captured explicitly by the model’s seasonal component. The forecast and its uncertainty bounds provide a statistically grounded basis for price risk assessment over the coming year.

The forecast should nonetheless be treated with appropriate caution. All three models are purely backward-looking and cannot anticipate the events that have historically caused the largest price dislocations, including disease outbreaks affecting Norwegian biomass, shifts in production licensing, and EUR/NOK exchange rate movements. The most productive direction for future work is to incorporate such exogenous drivers as additional inputs to the modeling framework, which would preserve the statistical rigour of ARIMA while adding forward-looking economic information that a purely historical time series structure cannot use.

References

- Andrews, D. W. K. (1993). Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61(4), 821–856.
- Asche, F., Misund, B., & Oglend, A. (2016). Determinants of the atlantic salmon futures risk premium. *Journal of Commodity Markets*, 2(1), 6–17.
- Asheim, L. J., Dahl, R. E., Kumbhakar, S. C., Oglend, A., & Tveteras, R. (2011). Are prices or biology driving the short-term supply of farmed salmon? *Marine Resource Economics*, 26(4), 343–357.
- Bloznelis, D. (2018). Short-term salmon price forecasting. *Journal of Forecasting*, 37(2), 151–169.
- Box, G. E. P., & Jenkins, G. M. (1970). *Time series analysis: Forecasting and control*. Holden-Day.
- Cleveland, R. B., Cleveland, W. S., McRae, J. E., & Terpenning, I. (1990). Stl: A seasonal-trend decomposition procedure based on loess. *Journal of Official Statistics*, 6(1), 3–33.
- Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366), 427–431.
- Granger, C. W. J. (1969). Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3), 424–438.
- Guttormsen, A. G. (1999). Forecasting weekly salmon prices: Risk management in fish farming. *Aquaculture Economics & Management*, 3(2), 159–166.
- Hurvich, C. M., & Tsai, C.-L. (1989). Regression and time series model selection in small samples. *Biometrika*, 76(2), 297–307.
- Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). OTexts. <https://otexts.com/fpp3/>
- Hyndman, R. J., & Khandakar, Y. (2008). Automatic time series forecasting: The forecast package for r. *Journal of Statistical Software*, 27(3), 1–22.
- Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4), 679–688.
- Hyndman, R. J., Koehler, A. B., Ord, J. K., & Snyder, R. D. (2008). *Forecasting with exponential smoothing: The state space approach*. Springer.
- Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., & Shin, Y. (1992). Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics*, 54(1–3), 159–178.
- Ljung, G. M., & Box, G. E. P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303.
- Lütkepohl, H. (2005). *New introduction to multiple time series analysis*. Springer.
- Norwegian Seafood Council. (2025). *2024 was the best year ever for norwegian seafood exports*. <https://en.seafood.no/news-and-media/news-archive/2024-was-the-best-year-ever-for-norwegian-seafood-exports/>
- Schwarz, G. (1978). Estimating the dimension of a model. *Annals of Statistics*, 6(2), 461–464.
- Sims, C. A. (1980). Macroeconomics and reality. *Econometrica*, 48(1), 1–48.
- Statistics Norway. (2026). *Salmon export statistics, table 03024*. <https://data.ssb.no/api/v0/en/table/03024>
- Tharmarajah, T., & Gjesdal, B. (2020). *Salmon price forecasting: A comparison of univariate and multivariate forecasting methods* [Master’s thesis]. Norwegian School of Economics. <https://hdl.handle.net/11250/2737169>
- Zeileis, A., Leisch, F., Hornik, K., & Kleiber, C. (2002). Strucchange: An r package for testing for structural change in linear regression models. *Journal of Statistical Software*, 7(2), 1–38.

Appendices

A. Seasonal and Subseries Plots

The seasonal and subseries plots below complement the discussion in Section 4. The seasonal plot displays each year separately and highlights the recurring within-year price pattern. The subseries plot shows the distribution of observations within each month across the full sample period.

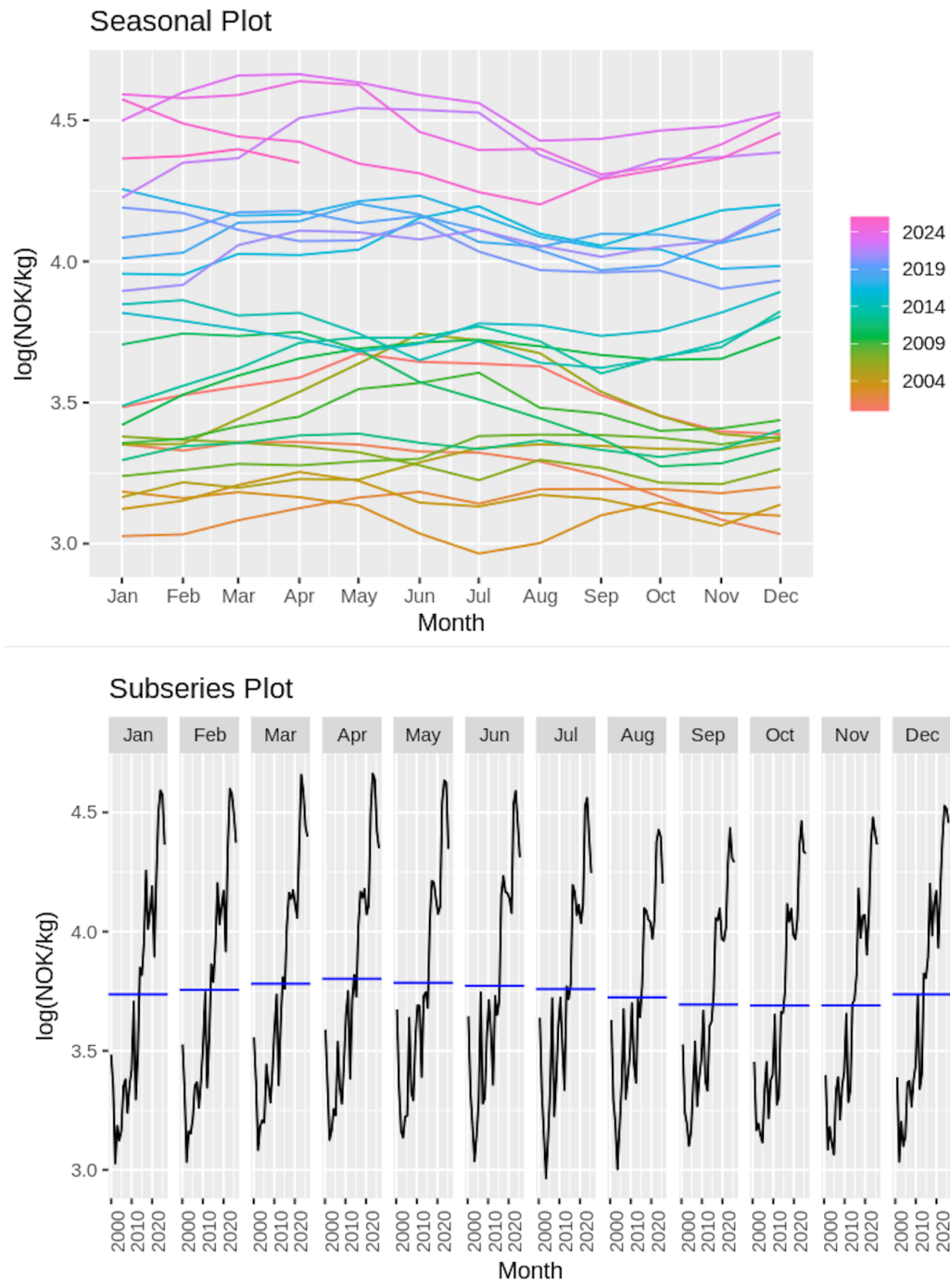


Figure 13: Seasonal plot (top) and subseries plot (bottom) of log salmon export price, January 2000 – April 2026.

B. Stationarity Tests

Table 3 reports the ADF and KPSS test results referenced in Section 4.3, both in levels and after one regular difference of the log price series.

Table 3: Stationarity test results for log salmon export price, in levels and after one regular difference. ADF rejects its null when the statistic falls *below* the critical value; KPSS rejects when the statistic lies *above* it.

Series	Test (H_0)	Statistic	1%	5%	10%	Decision
Levels	ADF-trend (unit root)	-4.066	-3.98	-3.42	-3.13	Reject $H_0^\dagger$
	KPSS- τ (stationary)	0.345	0.216	0.146	0.119	Reject H_0
First difference	ADF-none (unit root)	-4.493	-2.58	-1.95	-1.62	Reject H_0
	KPSS- μ (stationary)	0.063	0.739	0.463	0.347	Fail to reject H_0

[†] The ADF rejection in levels is marginal and coincides with a strong KPSS rejection of stationarity, indicating near-unit-root behaviour; first differencing is applied as the conservative remedy. After differencing, both tests agree the series is stationary, confirming an order of integration of one.

C. Additional Diagnostic Figures

The figures below provide enlarged versions of the diagnostic plots presented in the main text.

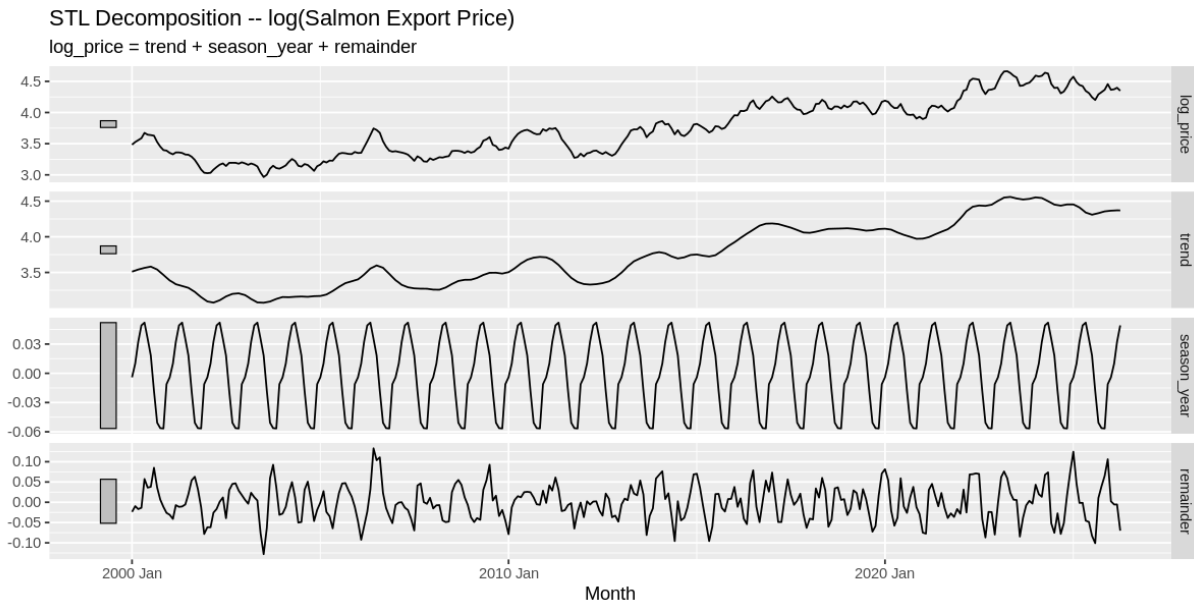


Figure 14: STL decomposition of log salmon export price. Trend window = 13 months, periodic seasonal window.

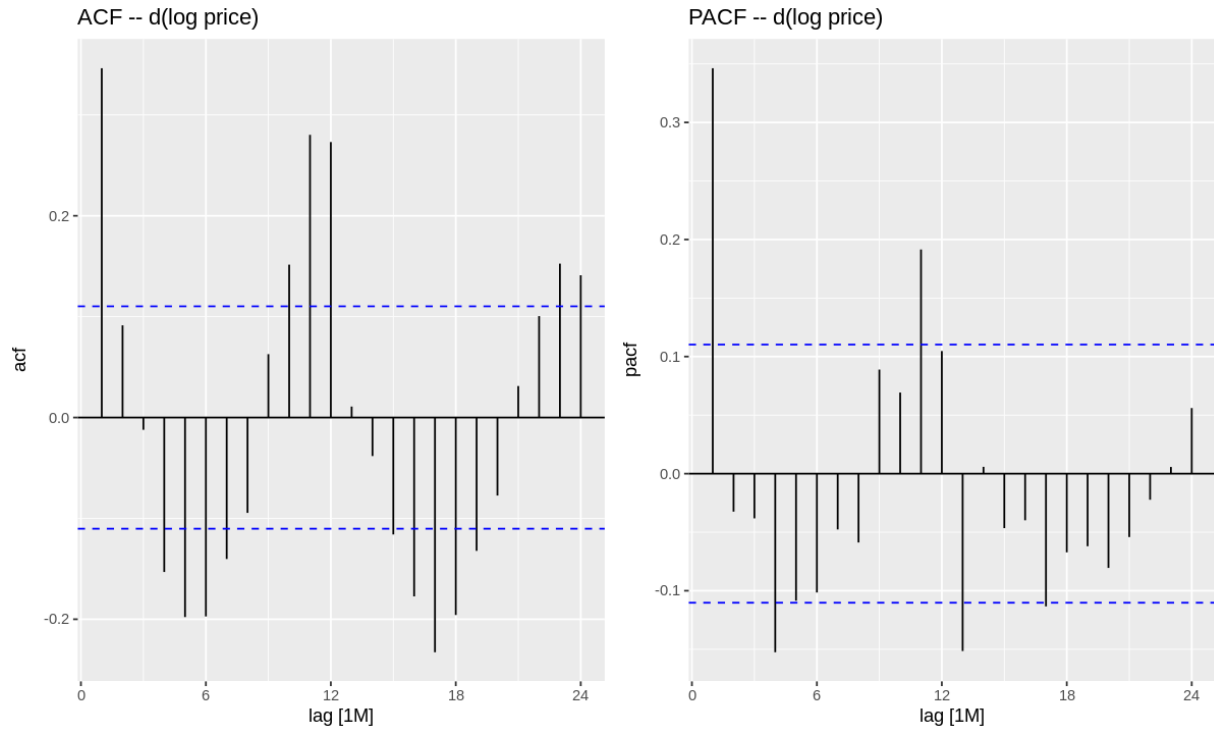


Figure 15: ACF and PACF of first-differenced log salmon export price.

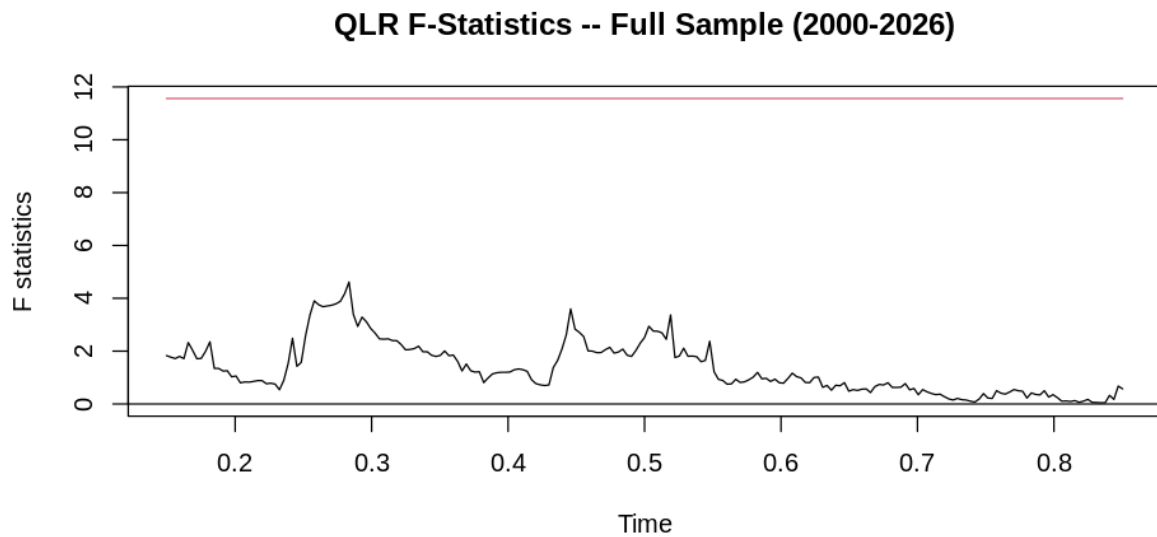


Figure 16: QLR F-statistics for the first-differenced log price series.

D. Model Selection

Table 4 compares the manual and automatic ARIMA specifications discussed in Section 5.2 on the training sample.

Table 4: ARIMA model selection on the training sample (January 2000 – December 2022). Final selection follows AICc (Hurvich & Tsai, 1989).

Model	$\log L$	σ^2	AIC	AICc	BIC
Manual: ARIMA(0,1,1)(1,0,0) ₁₂	442.54	0.00235	−879.09	−879.00	− 868.24
Auto: ARIMA(2,1,1)(1,0,0) ₁₂	446.96	0.00229	− 883.92	− 883.69	−865.83

Bold marks the best value for each criterion. The automatic search (Hyndman–Khandakar, `stepwise = FALSE`) adds two AR terms relative to the manual candidate, lowering AIC and AICc; the manual model attains marginally lower BIC due to its smaller parameter count. AICc is used for final selection given the sample size.

E. Residual Diagnostics

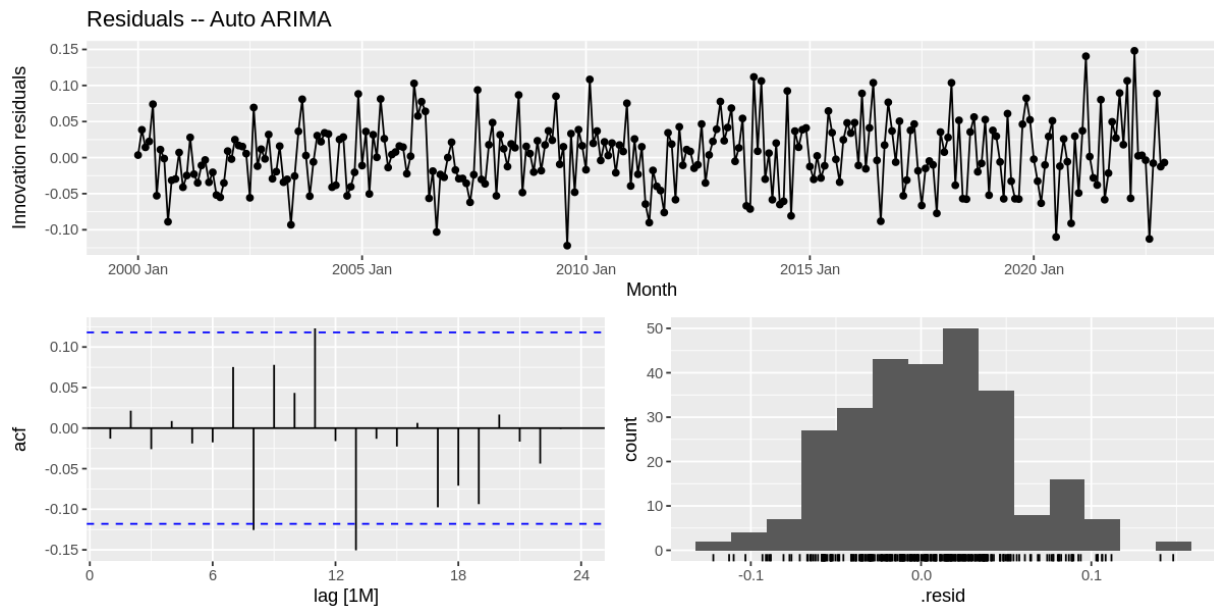


Figure 17: Residual diagnostics for ARIMA(2,1,1)(1,0,0)[12].

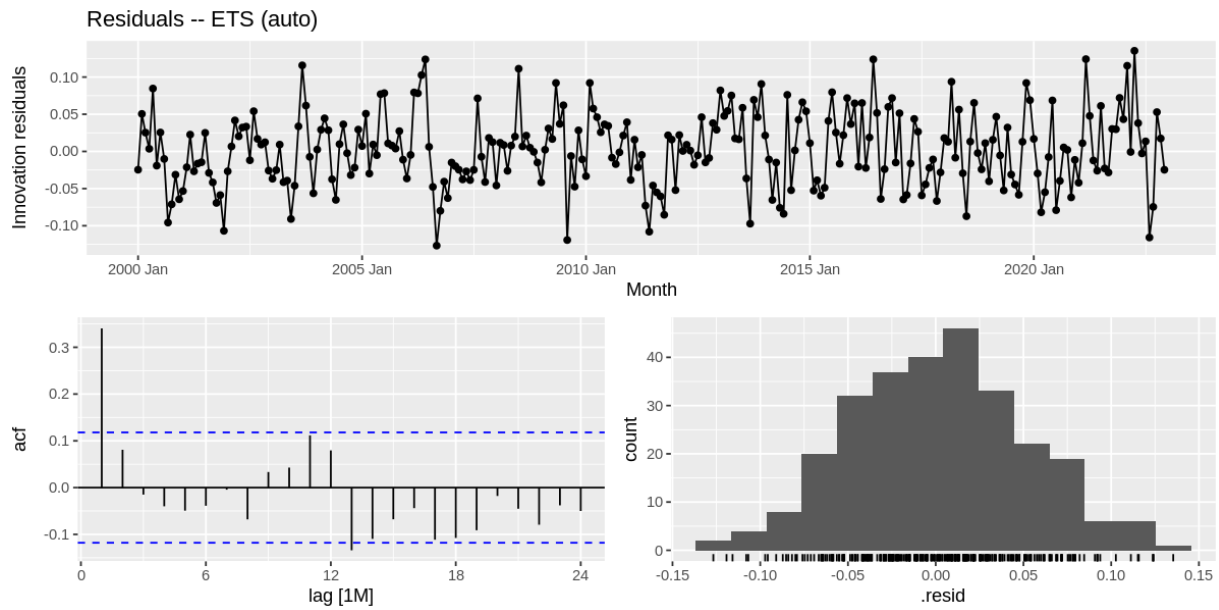


Figure 18: Residual diagnostics for ETS(A,A,A).

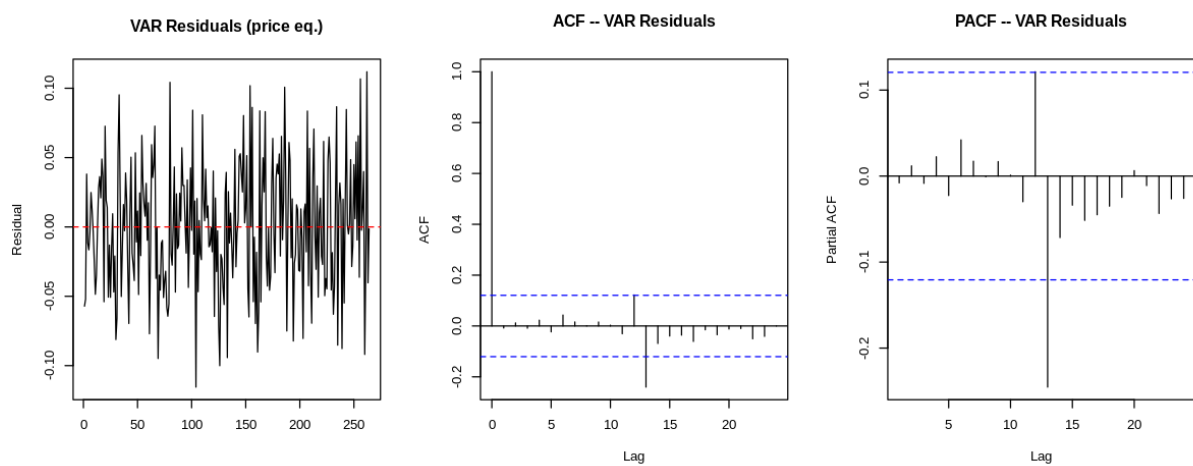


Figure 19: Residual diagnostics for VAR(11), price equation.

F. Forecast Figures

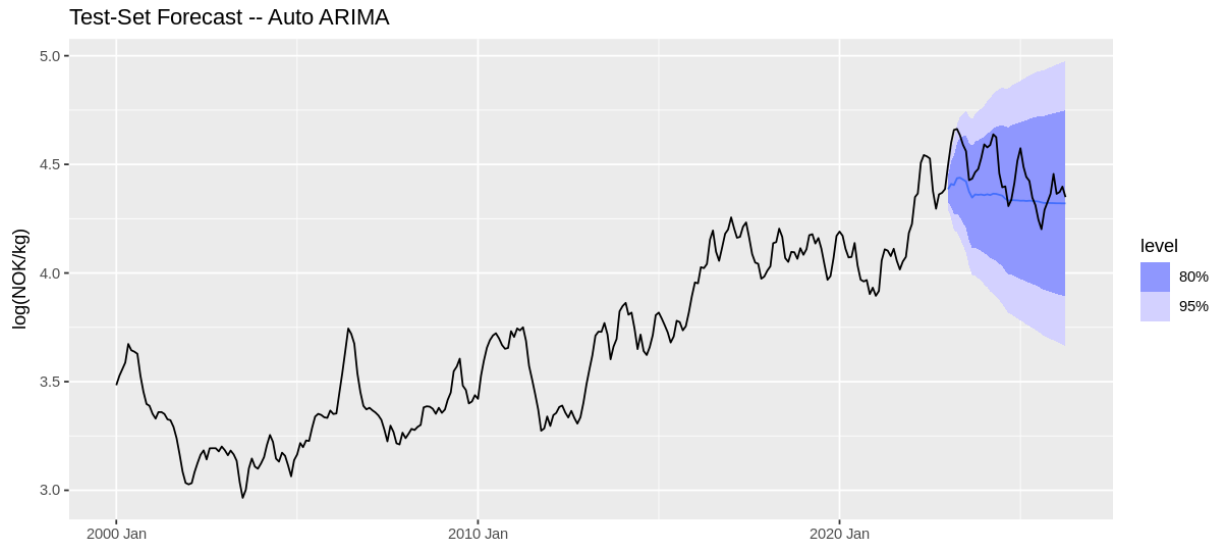


Figure 20: Test-set forecast for $\text{ARIMA}(2,1,1)(1,0,0)[12]$, January 2023 – April 2026.

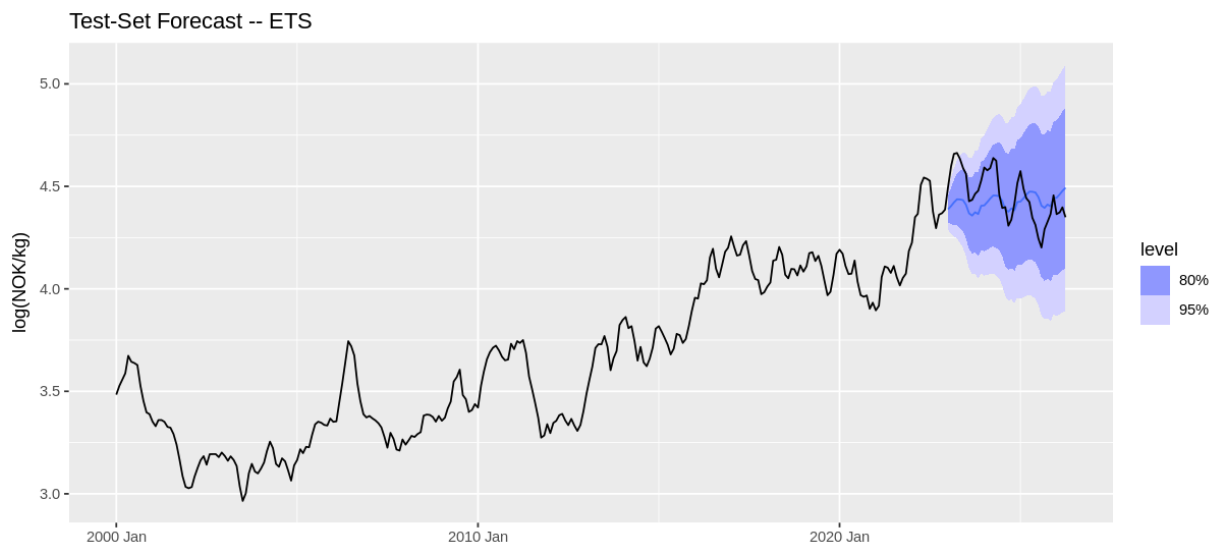


Figure 21: Test-set forecast for $\text{ETS}(A,A,A)$, January 2023 – April 2026.

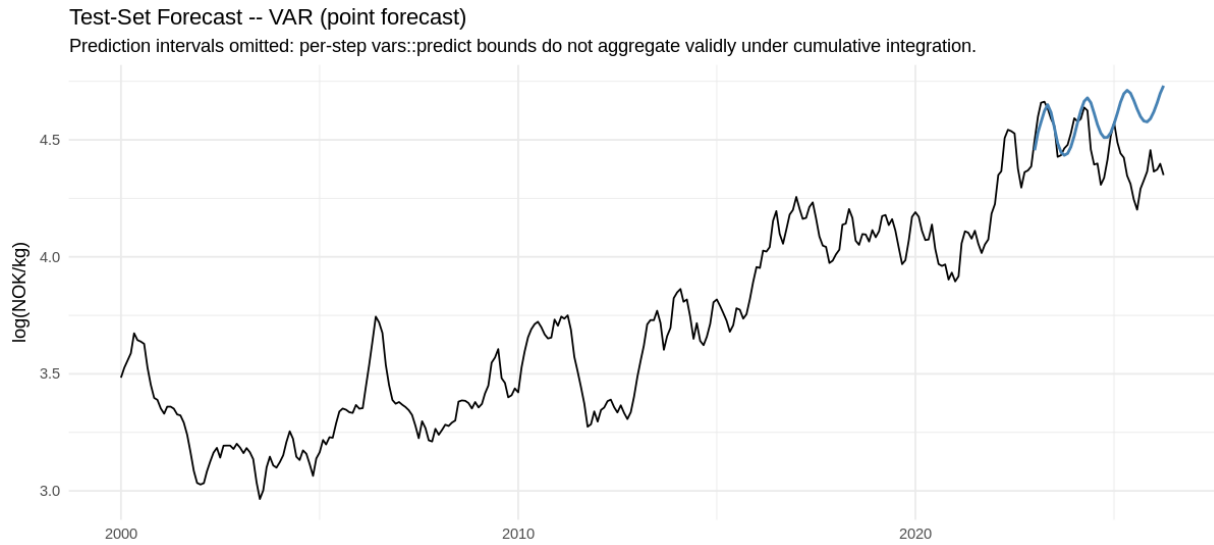


Figure 22: Test-set forecast for VAR(11), January 2023 – April 2026.

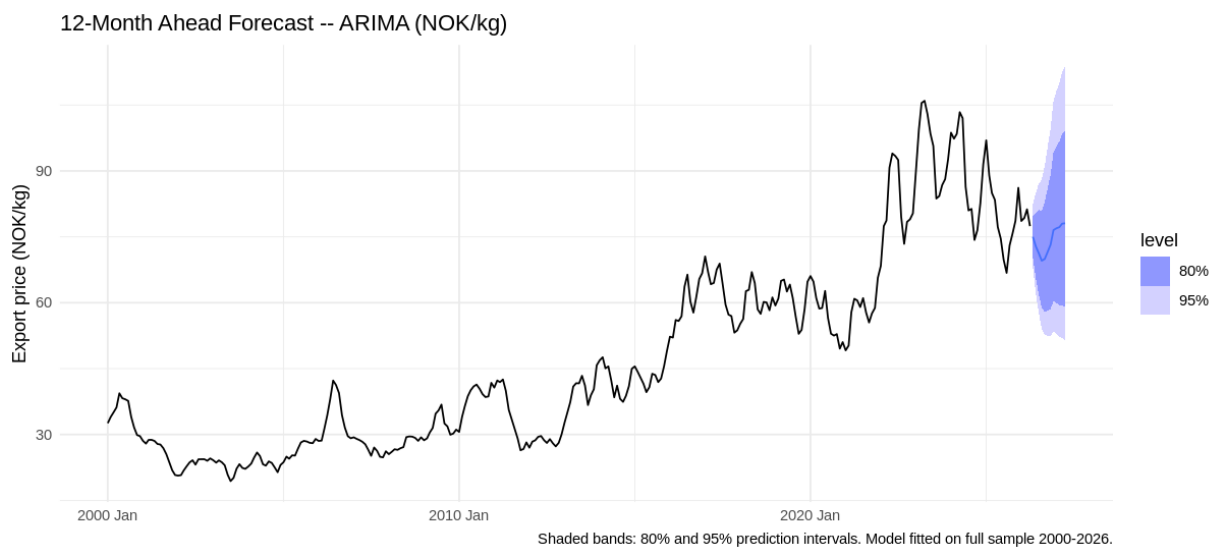


Figure 23: Final 12-month ARIMA forecast of Norwegian salmon export price, May 2026 – April 2027.